

LES GRANDES CHUTES, QC, Canada

WMO: 713910

Lat: 45.840N

Lon: 75.649W

Elev: 195

StdP: 99.01

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.4	-23.7	-30.5	0.2	-25.9	-27.8	0.3	-23.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	29.6	21.5	27.7	20.2	26.1	19.5	22.9	26.9	21.9	25.7	20.9	24.3	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.8	24.6	20.7	15.7	23.7	19.7	14.8	22.7	68.9	27.0	65.1	25.8	61.4	24.4	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-29.5	32.1	4.0	1.7	-32.4	33.3	-34.7	34.3	-36.9	35.3	-39.8	36.6
			-29.6	24.8	4.0	1.5	-32.5	25.9	-34.9	26.8	-37.1	27.6	-40.1	28.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-11.0	-9.5	-3.4	4.4	12.0	17.1	19.7	18.9	14.8	7.6	0.8	-6.5
	DBStd	11.85	7.26	6.31	6.09	4.57	4.16	3.52	2.81	2.87	3.93	4.35	4.67	6.26
	HDD10.0	2713	652	546	417	178	26	0	0	0	5	102	278	511
	HDD18.3	4842	911	779	675	419	201	64	17	28	119	334	527	769
	CDD10.0	1066	0	0	1	9	88	214	302	277	148	27	1	0
	CDD18.3	152	0	0	0	0	5	27	61	46	12	1	0	0
	CDH23.3	1167	0	0	0	4	75	256	454	296	81	1	0	0
	CDH26.7	274	0	0	0	1	17	66	114	61	17	0	0	0
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	991	70	58	64	73	76	93	97	97	97	90	83	84
	PrecMax	1237	108	121	125	131	129	170	178	140	164	158	150	141
	PrecMin	853	31	22	26	22	32	28	30	44	39	28	30	24
	PrecStd	91	21	26	29	30	28	36	40	24	36	33	27	30
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	6.1	13.3	22.8	29.0	31.5	32.0	30.8	29.0	22.2	14.1	8.4
		MCWB	5.3	4.1	7.5	14.5	19.5	22.4	23.7	21.4	21.8	17.1	11.6	7.7
	2%	DB	3.0	3.2	9.1	18.3	25.4	28.6	29.6	28.3	25.4	18.8	11.6	5.0
		MCWB	2.2	1.7	5.1	11.3	16.4	20.4	21.9	20.7	20.1	14.9	9.6	4.4
	5%	DB	1.1	1.6	6.7	14.9	22.4	26.3	27.7	26.5	23.2	16.3	9.2	2.8
		MCWB	0.5	0.5	3.3	8.9	14.5	18.9	20.5	19.9	18.5	13.2	7.6	2.1
	10%	DB	-0.5	0.2	4.5	12.1	19.9	24.2	25.9	24.8	21.4	14.1	7.3	1.2
		MCWB	-1.2	-0.8	2.0	6.9	13.3	17.8	19.8	19.1	17.8	11.8	5.8	0.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.4	4.4	9.5	15.6	21.2	24.1	25.1	23.5	22.8	18.2	12.7	7.8
		MCDB	5.5	5.3	11.7	20.5	26.9	29.1	30.3	27.1	26.4	20.5	13.5	8.1
	2%	WB	2.4	2.0	5.6	12.8	18.5	22.0	23.2	22.4	21.2	15.6	9.9	4.6
		MCDB	2.8	2.8	8.1	16.3	22.7	26.9	27.2	26.0	23.9	17.8	11.0	5.1
	5%	WB	0.6	0.7	3.7	9.8	16.5	20.5	22.0	21.3	19.8	13.7	7.9	2.2
		MCDB	1.1	1.4	6.1	13.5	19.6	24.3	25.6	24.6	22.4	15.8	9.0	2.7
	10%	WB	-1.2	-0.8	2.3	7.8	14.6	19.1	20.9	20.3	18.4	12.1	5.9	0.6
		MCDB	-0.5	0.3	4.2	11.5	18.3	22.3	24.4	23.3	20.8	14.0	7.1	1.1
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	10.4	10.2	10.1	11.0	10.6	10.5	10.2	9.9	7.8	6.6	6.9
		MCDBR	8.2	9.1	11.7	16.0	16.0	14.3	13.0	12.4	12.1	11.3	9.4	6.7
	5% WB	MCWBR	7.9	8.1	8.0	9.3	7.6	7.0	6.2	5.9	7.0	7.3	7.8	6.6
		MCWBR	8.2	8.5	10.5	13.9	12.6	11.8	11.1	10.4	10.8	10.4	8.9	6.7
(40) Clear-Sky Solar Irradiance	taub	taub	0.255	0.280	0.321	0.337	0.374	0.401	0.403	0.387	0.363	0.331	0.305	0.272
		taud	2.363	2.273	2.302	2.427	2.369	2.305	2.307	2.364	2.417	2.509	2.500	2.369
	Edn at Noon	Edn at Noon	864	899	907	923	893	866	859	861	853	836	798	795
		Edh at Noon	72	97	111	107	118	127	125	114	99	78	65	66
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.37	2.34	3.70	4.55	5.15	5.63	5.65	4.98	3.88	2.27	1.36	0.99
	RadStd	RadStd	0.18	0.26	0.33	0.42	0.50	0.42	0.49	0.24	0.33	0.21	0.17	0.09

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page